# The Generative Incompleteness

**Graziano Guiducci · Revised manuscript · 28 October 2026**

*One recurrence. An invariant direction. A continuing state.*

The D-ND source describes the movement of possibility through an indeterminate reference and a dipolar relation that contains its poles. The observer is the resultant entity of alignment; observer, alignment and resultant are described within the same movement. Mathematics is included in this wider logical model. Paper Zero presents one particular mathematical representation: a rational recurrence with an irrational attracting projective direction.

The calculation makes a precise relationship inspectable: a direction can remain invariant while the state that carries it continues to change.

## The map

$$f(x) = 1 + \frac{1}{x}.$$

For a positive starting value, every affine iterate is defined and the sequence converges to the golden ratio. The table also includes a negative starting value whose particular orbit remains defined. Values are rounded to three decimal places.

| Step | $x_0 = 7$ | $x_0 = 0.1$ | $x_0 = -3$ |
|---|---|---|---|
| 0 | 7.000 | 0.100 | -3.000 |
| 1 | 1.143 | 11.000 | 0.667 |
| 2 | 1.875 | 1.091 | 2.500 |
| 3 | 1.533 | 1.917 | 1.400 |
| 4 | 1.652 | 1.522 | 1.714 |
| 5 | 1.605 | 1.657 | 1.583 |
| 6 | 1.623 | 1.603 | 1.632 |
| 7 | 1.616 | 1.624 | 1.613 |
| 8 | 1.619 | 1.616 | 1.620 |

The two real fixed points are

$$\varphi = \frac{1+\sqrt{5}}{2}, \qquad \psi = \frac{1-\sqrt{5}}{2} = -\frac{1}{\varphi}.$$

On the real projective line, the transformation is defined at every point and converges to the direction represented by $\varphi$ from every initial direction except the fixed repeller $\psi$. In the affine chart, zero and its finite preimages encounter division by zero: for example, $-1 \mapsto 0$ has no next affine value. The projective orbit continues through the point at infinity.

## The matrix and the convergence

$$M = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \operatorname{tr} M = 1, \qquad \det M = -1.$$

The characteristic polynomial is $\lambda^2 - \lambda - 1$. Its eigenvalues are $\varphi$ and $\psi$. Both are quadratic algebraic irrationals.

Let $r = (\varphi, 1)^T$ and $s = (\psi, 1)^T$. Every nonzero vector has the form $v = ar + bs$. For $a \neq 0$,

$$[M^n v] = \left[ r + \frac{b}{a} \left( \frac{\psi}{\varphi} \right)^n s \right] \longrightarrow [r], \qquad \left| \frac{\psi}{\varphi} \right| = \varphi^{-2} < 1.$$

Brackets denote a projective direction: nonzero scalar multiples represent the same point. If $a = 0$, the direction is $[s]$ and remains fixed. This proves the stated projective convergence, including its exception.

Finite rational affine iterates remain rational whenever defined. Their attracting limit is irrational. The title names this relationship between the rational field of the finite iterates and the invariant direction in its quadratic extension $\mathbb{Q}(\sqrt{5})$. This is the algebraic statement established here. Gödel's incompleteness theorems concern provability in formal theories under their own hypotheses.

## Orientation and growth

The determinant records an exact geometric property: for any two vectors, the signed area they span changes sign under $M$, while its magnitude is preserved.

Growth and periodicity also depend on the eigenstructure. For comparison,

$$B = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad B^2 = I, \quad \det B = -1; \qquad C = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \quad \det C = 1.$$

$B$ is periodic; $C$ has expanding eigenvalue $(3 + \sqrt{5})/2$. Within the chosen reciprocal pair, changing the rule to $g(x) = 1 - 1/x$ gives

$$N = \begin{pmatrix} 1 & -1 \\ 1 & 0 \end{pmatrix}, \qquad N^3 = -I.$$

Its projective transformation has period three. This comparison distinguishes these two rules. The sign of the determinant alone does not classify generative growth.

## Why the golden ratio appears

The positive fixed-point equation $x = 1 + 1/x$ gives $x^2 - x - 1 = 0$ and therefore $x = \varphi$. Its continued fraction is $[1; 1, 1, 1, \ldots]$.

A precise minimality statement is available: among primitive, nonnegative integer $2 \times 2$ matrices with determinant $-1$, the smallest spectral radius is $\varphi$. Indeed, their integer trace $t$ is at least one, and the spectral radius is $(t + \sqrt{t^2 + 4})/2$. At $t = 1$, the diagonal entries are zero and one, and the off-diagonal entries are both one. The two representatives are $M$ and $PMP$, where $P = B$ exchanges the coordinates.

Entropy belongs to a specified dynamical system. With $M$ as an adjacency matrix, the golden-mean shift has topological entropy $\log \varphi$ (natural logarithm). The Fibonacci substitution with the same incidence matrix has entropy zero. The projective map studied here is a circle homeomorphism and also has entropy zero. These are different uses of the same matrix.

## A fixed direction and a continuing state

At the attracting direction,

$$Mr = \varphi r, \qquad M^n r = \varphi^n r, \qquad [M^n r] = [r].$$

The vector changes scale at every step; its projective direction stays fixed. This relationship was already developed in the D-ND manuscript on relational distinction and continuity of the resultant, Appendix B, 20 October 2026.

Within this representation, the expression $R + 1 = R$ can be read as one further application of the projective transformation leaving its fixed direction invariant: $T([r]) = [r]$. This is one mathematical reading of the D-ND resultant. The source meaning of resultant, reference, movement and observer remains available beyond this chosen representation.

## The two coordinates

Write the state as a related pair of coordinates $(D, ND)$. Then

$$(D, ND) \longmapsto (D + ND, D), \qquad x = \frac{D}{ND}, \qquad x' = \frac{D + ND}{D} = 1 + \frac{1}{x}.$$

Both coordinates are already present in the input and are updated in one operation. Their names provide the interpretation chosen for this construction; the dipolar relation that contains its poles belongs to the wider D-ND source. The determinant describes area, rather than conservation of the coordinate product $D \cdot ND$.

## The numerical first step

From $x_0 = 7$, the first iterate is exactly $8/7$ and the fourth is $38/23$. Relative to the initial absolute error $|7 - \varphi|$, the error falls by approximately **91.170943%** after one step and **99.365661%** after four steps. The familiar rounded values, **91.2%** and **99.4%**, express this numerical example. The percentages measure reduction of absolute error.

## The relationship made visible

An integer matrix has an irrational attracting eigen-direction. Rational projective iteration approaches that direction on the stated domain; an eigenvector on it continues to change scale. Paper Zero makes this relation of invariance and continuation explicit within the wider D-ND research.

The algebraic proof is given above. Iteration makes its consequence observable.

## Sources and edition

The D-ND model and its source observations are attributed to Graziano Guiducci. The authorial clarification of 27 October 2026 places mathematics within the general logical model. This revised manuscript uses classical mathematics; its mathematical examples and interpretations are distinguished from those authorial formulations.

For the entropy distinction: H. Bruin, *Ergodic Theory I*, Lecture 18, University of Vienna; M. Baake, T. Spindeler and N. Strungaru, *Diffraction of compatible random substitutions in one dimension*, Indagationes Mathematicae 29 (2018), 1031–1071, DOI: 10.1016/j.indag.2018.05.008. The projective entropy statement also follows directly from the circle-partition argument recorded in the accompanying revision notes.

Historical edition: March 2026, DOI **10.5281/zenodo.18902950**, the archived repository release v2.0.0. That DOI identifies the historical deposit; this revised manuscript has its own dated identity.